Proposal: Stroke Prediction with Class Imbalance: Model Comparison and Handling Strategies

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1. What research question are you trying to answer?

R1: Which machine learning algorithms trained on the Kaggle Stroke Prediction Dataset best predict whether a patient is likely to have a stroke?
R2: Which features are most important for accurate prediction of stroke occurrence?
R3a: Do existing methods for handling class imbalance improve the accuracy of predictions for individuals that *are* likely to have a stroke (i.e., predictions for the minority class) compared to baseline models? If so, which methods result in the greatest performance increase?
R3b: Can we design imbalance-handling methods tailored to this dataset's characteristics that match or improve on general methods? How might such methods inform the design of models for similar imbalanced settings?

2. Why is this question interesting to you?

Real world data is often messy, and contains instances of class imbalance. Models naively trained on imbalanced data may fall victim to pitfalls such as only predicting the majority class, which may result in high accuracy, but low F1. The Kaggle Stroke Prediction Dataset exhibits such class imbalance, with 95% of patients represented in the data having no stroke, and only 5% having a stroke. We are thus interested in exploring the impact of class imbalance on the predictive power of different machine learning models on this dataset. We are also interested in exploring ways to account for class imbalance and increase the accuracy of these models, particularly on the minority class. The capability to gracefully handle class imbalance is critical in situations such as prediction of whether a patient is likely to have a stroke. False negatives could prevent patients from seeking the appropriate preventative care, or making necessary lifestyle changes to minimize their risk of stroke.

3. What kind of data are you collecting or what datasets will you use?

We plan to use the Kaggle Stroke Prediction Dataset [1] for this project. This dataset has 10 features for each patient: gender, age, hypertension, heart disease, whether the patient has ever been married, work type, residence type, average glucose level of the patient, BMI, and smoking status. Gender, whether the patient has ever been married, work type, residence type, and smoking status are all categorical features. While hypertension and heart disease may also be considered categorical, they appear naturally in the dataset as a binary feature. Age, average glucose level, and BMI are all numerical features. Each patient also has a label which indicates whether they have had a stroke event. We will use the 10 features to predict whether the patient is likely to have had a stroke.

This dataset passes certain sanity checks that indicate it is likely to be real rather than synthetically generated. For example, it does not contain young children who are married, smoke, or have a non-child work type. Additionally, Kaggle reports that this data comes from patient records, and the author of this dataset is fedesoriano, a datasets grandmaster at Kaggle. We believe this indicates that this dataset is likely reliable and legitimate.

4. What algorithms will you try?

Our choice of baseline models are informed by prior work with this dataset. Specifically, we consider algorithms such as logistic regression, SVM, Random Forests, XGBoost, K-NN, Naive Bayes, and decision trees as initial candidates for the baseline models, as these have previously been used on this data with moderate success [2, 3, 4]. We will leverage scikit-learn's implementation of these algorithms as the basis for our code. We also plan to apply at least one type of neural network to this problem.

To address the severe class imbalance in the dataset (≈95% no stroke, 5% stroke), we will apply and compare four standard approaches in addition to our baseline models.

(1) Class weight : We will use per-class loss weights (e.g. scikit-learn's class_weight='balanced') so that misclassifying the minority class is penalized more heavily, without changing the number of samples [5].
(2) Random oversampling : We will balance the training set by resampling or duplicating minority-class examples so that the model sees more stroke-positive cases [6].
(3) SMOTE (Synthetic Minority Over-sampling Technique) : We will use SMOTE to generate synthetic minority samples in feature space (e.g. via k-NN interpolation) rather than simple duplication, and train on the resampled data [6].
(4) Using a confidence score: Since there is likely to be a decent amount of randomness in whether someone has a stroke, our predictor can output a *confidence score* between 0 and 1 indicating the *probability* of each patient having a stroke, perhaps postprocessing the model as necessary to ensure calibration and perhaps multicalibration [7, 8].

We will evaluate each approach with the same metrics (precision, recall, F1) and compare them to our baselines to answer R3.

5. What experiments and analysis will you run?

As our dataset only contains a single split, we plan to randomly partition it into an 80/10/10 train/validation/test split. To answer R1, we plan to train our baseline models on all features and evaluate them via standard metrics such as accuracy, precision, recall, F1 on the appropriate splits. Similar to prior work, at minimum we plan to explicitly report precision, recall, and F1, as these are more sensitive to the effects of class imbalance than accuracy [2, 3, 4]. To answer R2, we will consider the impact of existing feature selection methods, such as l-1 regularization, relative to each of the baseline models. We will train new models using feature selection. We will choose evaluation metrics that allow direct comparison to the respective baseline models, and will additionally report the features selected for each model. To answer R3, we will consider the impact of existing and new methods of handling class imbalance relative to each baseline model. Similarly, we will report the standard evaluation metrics relative to the baseline, as well as any metrics or outcomes associated with the imbalance method under consideration.

6. What do you plan to finish by the pre-final report and check-in?

By the week of April 20th, we plan to have completed our experiments for R1, R2, and R3 for the models that leverage scikit-learn's implementations. Ideally, we will have also completed our experiments for R1, R2, and R3 for the neural networks we use; however, we may only have a subset of the research questions for the neural networks answered by April 20th, depending on the complexity of the implementation.

We plan to maintain documentation of our experiments and results for each combination of model, feature selection, and imbalance approach. Thus, by the week of April 20th, we should also have an informal document and figures that present our initial conclusions about which models performed best, which features were most relevant, and which imbalance approaches were most effective. We should be able to present initial conclusions for the scikit-learn-implemented models for all research questions at this stage, and a subset of the conclusions for neural networks. We should also have the ability to present initial conclusions about how our experiments might inform future work on similar real-world datasets with severe class imbalance.

References

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